UNIGUARD DIGITAL WALLET

AI-Driven Personal Budgeting and Financial Literacy Enhancement for University Students

"Turning Pocket Money into Smart Money"

UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

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ABSTRACT

Financial illiteracy and poor budgeting habits among youth and students remain persistent global challenges
This research project focuses on helping university students and young people manage their money better. Many students depend on small or irregular sources of income, such as pocket money, allowances, gifts, or part-time jobs. Because of this, they often struggle to plan their spending, save money, or avoid overspending. Many also lack financial knowledge, which makes the problem worse.

INTRODUCTION

UniGuard Wallet is a user-friendly mobile app created specifically for university students in Uganda who often face money challenges. Most rely on unpredictable income from allowances or small gigs, but without proper budgeting skills or financial know how, they end up overspending on things like airtime and outings. As noted by the National Council for Higher Education (2018), this frequently causes debt, anxiety, and lower grades. UniGuard makes it simple with instant spending tracking, smart AI that builds realistic budgets for irregular income, enjoyable games that teach money management, and notifications for campus jobs, scholarships, and deals. It helps students take control, spend smarter, feel less stressed, develop strong habits, and stay focused on their studies so they can finish university confident and financially secure.

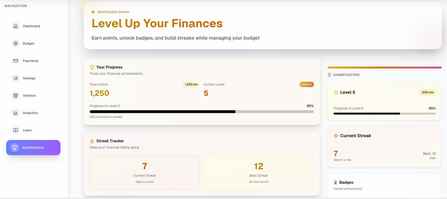
PROBLEM STATEMENT

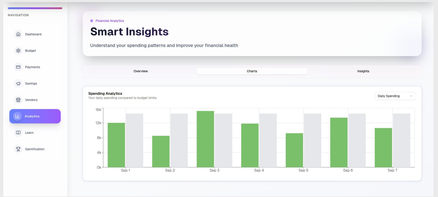
University students in Uganda face severe financial challenges due to low financial literacy (only 34-38% of Ugandans are financially literate – S&P Global FinLit Survey 2015; InterMedia FII), poor budgeting habits, and dependence on irregular income from allowances and casual gigs. This leads to impulse spending, mounting debt, high stress levels, and academic underperformance (National Council for Higher Education, 2018). The lack of accessible, student-focused financial tools creates a critical gap, preventing young people from building responsible money habits during their university years.

OBJECTIVES

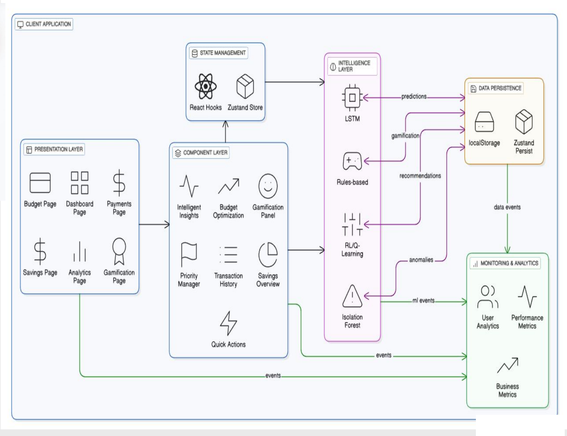
- **Develop a smart budgeting tool tailored to students (Web Application).**
- **Provide AI-driven budget optimisation and overspending prediction.**
- **Deliver interactive financial literacy gamification.**
- **Build a secure and lightweight mobile system.**

UNIGUARD DASHBOARD

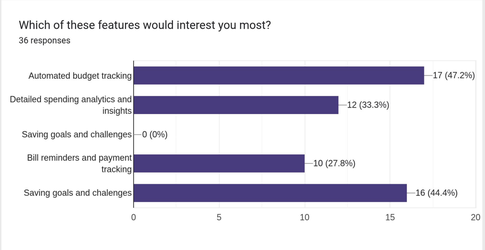


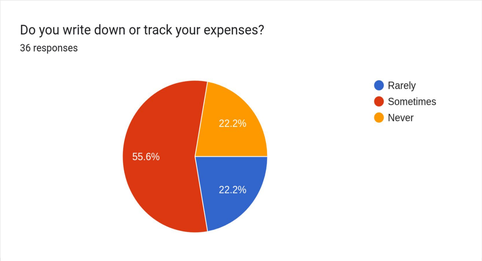


SYSTEM ARCHITECTURE



PRELIMINARY RESULTS





CONCLUSION:

UniGuard Wallet offers a practical and empowering solution to the financial struggles faced by university students, especially those relying on unpredictable income. By combining smart budgeting tools, real-time tracking, and engaging financial learning features, it equips users with the skills and discipline needed to manage money confidently. Ultimately, UniGuard helps students build lasting financial stability, reduce stress, and stay focused on achieving academic success.

PROPOSED SOLUTION

- **Smart Budgeting System: Track income types, categorize expenses, set spending limits**
- **Budget Optimization Engine: Distinguish essential vs. non-essential expenses, suggest weekly budgets..**
- **Financial Literacy Gamification: Offer challenges, rewards, leaderboards, and daily quizzes.**
- **Enhanced Security Layer: 2FA**

Alignment with SDGs & NDPIV

SDG 1 – No Poverty:

SDG 4 – Quality Education:

SDG 8 – Decent Work & Economic Growth

National Development Plan (NDP IV): Aligns with the Digital Economy Program and Human Capital Development.

FUTURE WORKS

- **Integration with mobile money APIs for automatic transaction syncing.**
- **Deploying reinforcement learning for adaptive budgeting.**
- **Advanced behavioural analytics dashboards.**
- **Beta testing with UCU students.**
- **Expansion to full cross-university deployment.**

REFERENCES

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